

SAT Unit 2.3.2

Exponential Functions

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Overview

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- 2 Graphs and Transformation
- 3 Application
- 4 Practices

Definition and Properties

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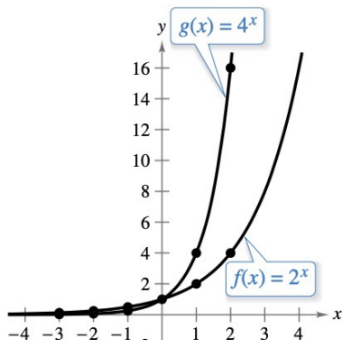
Definition of Exponential Function

Exponential Function: A function of the form:

$$f(x) = a^x$$

where:

- $a > 0$
- $a \neq 1$
- x is any real number



Key Properties of Exponential Functions

If $f(x) = a^x$, then:

- Domain: $\mathbb{R}$ (all real numbers)
- Range: $(0, \infty)$
- $f(0) = 1$ (since $a^0 = 1$)
- Continuous and smooth curve
- Horizontal asymptote at $y = 0$

Growth vs Decay:

- If $a > 1$, function exhibits **exponential growth**
- If $0 < a < 1$, function exhibits **exponential decay**

Question: Identifying k in an Exponential Function

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

Options:

- A. $f(x) = 50(2)^{x+1}$
- B. $f(x) = 80(2)^x$
- C. $f(x) = 128(2)^{x-1}$
- D. $f(x) = 205(2)^{x-2}$

Answer and Explanation

We are told $f(1) = k$. Let's plug $x = 1$ into each option and see which form gives $f(1) = k$ as the coefficient or base.

- **A.** $f(1) = 50(2)^2 = 50 \cdot 4 = 200$
- **B.** $f(1) = 80(2)^1 = 160$
- **C.** $f(1) = 128(2)^0 = 128 \cdot 1 = 128$
- **D.** $f(1) = 205(2)^{-1} = \frac{205}{2} = 102.5$

Only in option **C**, we get $f(1) = 128$, and the coefficient **is** k directly.

Correct Answer: C

Question: Exponential Growth Relationship

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

Options:

- A. $y = 4(x)^{200}$
- B. $y = 4(200)^x$
- C. $y = 200(2)^x$
- D. $y = 200(4)^x$

Answer and Explanation

This is an exponential relationship where y increases by a factor of 4 for every 1 increase in x . The general exponential form is:

$$y = a \cdot b^x$$

Given:

- Initial value: $y = 200$ when $x = 0$
- Growth factor: $b = 4$

So the equation becomes:

$$y = 200(4)^x$$

Correct Answer: D

Question: Finding Constant in Exponential Function

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has:

- an x -intercept at $(2, 0)$
- a y -intercept at $(0, -323)$

What is the value of b ?

Answer and Explanation

We are given two points:

- $f(2) = 0 \Rightarrow a^2 + b = 0$
- $f(0) = -323 \Rightarrow a^0 + b = -323$

From the second equation:

$$1 + b = -323 \Rightarrow b = -324$$

Substitute $b = -324$ into the first equation:

$$a^2 - 324 = 0 \Rightarrow a^2 = 324 \Rightarrow a = 18$$

Final Answer: $b = \boxed{-324}$

Graphs and Transformation

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Graphs of Exponential Functions

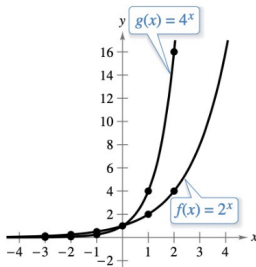
Graphs of $y = a^x$

In the same coordinate plane, sketch the graph of each function.

① $f(x) = 2^x$

② $g(x) = 4^x$

Begin by constructing a table of values.

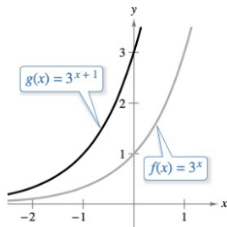


Observation:

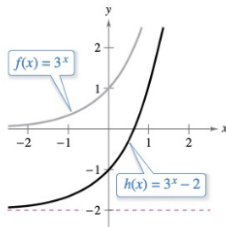
- Exponential functions increase/decrease rapidly
- All graphs pass through the point (0, 1)

Transformations of the Exponential Function

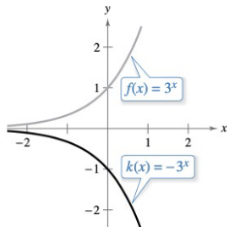
a.



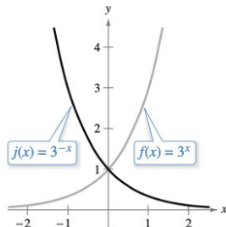
b.



c.



d.



Transformations of the Exponential Function

a. $g(x) = 3^{x+1}$

Since $g(x) = 3^{x+1} = f(x+1)$, the graph of g is a **horizontal shift left** by **1 unit**.

b. $h(x) = 3^x - 2$

Since $h(x) = 3^x - 2 = f(x) - 2$, the graph of h is a **vertical shift downward** by **2 units**.

c. $k(x) = -3^x$

Since $k(x) = -3^x = -f(x)$, the graph of k is a **reflection across the x -axis**.

d. $j(x) = 3^{-x}$

Since $j(x) = 3^{-x} = f(-x)$, the graph of j is a **reflection across the y -axis**.

Question: X-Intercepts of an Exponential Function

Given the equation:

$$y = 9 \left(\frac{a}{6} \right)^{x+c} - b,$$

how many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 6$ and $b > c$?

Options:

- A. Zero
- B. One
- C. Two
- D. Three

Answer and Explanation

We are given the exponential function:

$$y = 9 \left(\frac{a}{6} \right)^{x+c} - b$$

- Since $a > 6$, $\frac{a}{6} > 1$, meaning the function is **increasing**.
- The expression $9 \left(\frac{a}{6} \right)^{x+c}$ is always positive.
- As $x \rightarrow -\infty$, $\left(\frac{a}{6} \right)^{x+c} \rightarrow 0$, and $y \rightarrow -b$, so the horizontal asymptote is $y = -b$.
- As x increases, y increases beyond $-b$.
- Since $y = 0$ when $9 \left(\frac{a}{6} \right)^{x+c} = b$, and this equation has exactly one solution for x , the graph crosses the x -axis exactly once.

Correct Answer: B. One

Application

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Applications of Exponential Functions

Exponential functions are used to model real-world situations like:

- **Population Growth:** $P(t) = P_0 \cdot a^t$
- **Compound Interest:** $A = P(1 + r)^t$
- **Radioactive Decay:** $M(t) = M_0 \cdot a^{-t}$
- **Bacterial Growth**

Example: A population doubles every 3 years:

$$P(t) = 100 \cdot 2^{t/3}$$

Exponential Function with Time Units — Question

The given function $p(t) = 90,000(1.06)^t$ models the population of Lowell t years after a census.

Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \frac{90,000}{12}(1.06)^m$

B. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^m$

C. $r(m) = 90,000 \left(\frac{1.06}{12} \right)^{\frac{m}{12}}$

D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

Exponential Function in Terms of Different Units

The original exponential model is:

$$p(t) = 90,000(1.06)^t$$

where t is measured in **years**.

To express the model in terms of **months**, we must use:

$$t = \frac{m}{12}$$

Substitute into the original function:

$$r(m) = 90,000(1.06)^{\frac{m}{12}}$$

This gives us the population model in terms of months.

Correct Answer: D

$$r(m) = 90,000(1.06)^{\frac{m}{12}}$$

Exponential Modeling — Question

The function P models the population, in thousands, of a certain city t years after 2005:

$$P(t) = 290(1.04)^{\left(\frac{4}{6}\right)t}$$

According to the model, the population is predicted to increase by $n\%$ every 18 months.

What is the value of n ?

Choices:

- A. 0.38
- B. 1.04
- C. 4
- D. 6

Exponential Modeling — Solution

Given:

$$P(t) = 290(1.04)^{\left(\frac{4}{6}\right)t}$$

Step 1: Convert 18 months to years

$$18 \text{ months} = \frac{18}{12} = 1.5 \text{ years}$$

Step 2: Plug in $t = 1.5$ to evaluate the growth over 18 months

$$P(1.5) = 290(1.04)^{\left(\frac{4}{6}\right)(1.5)} = 290(1.04)^1$$

Conclusion: The population grows by a factor of 1.04 every 18 months, or **4%**.

Answer: C. $n = 4$

Practices

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Question

The function

$$f(t) = 20 \left(1 + \frac{15}{100} \right)^t$$

models the temperature, in degrees Celsius, of a metal alloy used in an experiment, where t is the number of seconds after the experiment began.

Which of the following is the best interpretation of the number 15 in this context?

- A) The temperature, in degrees Celsius, of the metal alloy at the beginning of the experiment
- B) The increase in the temperature, in degrees Celsius, of the metal alloy every 100 seconds during the experiment
- C) The percent by which the temperature, in degrees Celsius, of the metal alloy decreased from each second to the next during the experiment
- D) The percent by which the temperature, in degrees Celsius, of the metal alloy increased from each second to the next during the experiment

Answer

The function is:

$$f(t) = 20 \left(1 + \frac{15}{100} \right)^t = 20(1.15)^t$$

This is an exponential growth function of the form:

$$f(t) = a(1 + r)^t$$

Where $a = 20$ is the initial value, and $r = 0.15$ or 15% is the percent increase per second.

Thus, the number 15 represents the **percent increase per second** in temperature.

Correct answer: D

Question

The population of trees in a forest has been decreasing by 6 percent every 4 years. The population at the beginning of 2015 was estimated to be 14,000.

If P represents the population of trees in the forest t years after 2015, which of the following equations best defines P ?

A) $P = 14,000(0.06)^{\frac{t}{4}}$

B) $P = 14,000 + 0.94(4t)$

C) $P = 14,000(0.94)^{4t}$

D) $P = 14,000(0.94)^{\frac{t}{4}}$

Answer

We are told the population decreases by 6% every 4 years.
This is exponential decay, so the decay factor is:

$$1 - 0.06 = 0.94$$

Every 4 years, the population is multiplied by 0.94. Since t is in years, and 0.94 applies every 4 years, the exponent must be $\frac{t}{4}$.

So the equation is:

$$P = 14,000(0.94)^{\frac{t}{4}}$$

Correct answer: D

Question

The amount of data, in gigabytes, stored in a database increases by 2% every 15 hours. If 16 gigabytes worth of data are currently stored in the database, which of the following functions g gives the amount of data, in gigabytes, that will be stored in the database t days from now?

A) $g(t) = 16(2)^{15t}$

B) $g(t) = 16(1.02)^{\frac{t}{15}}$

C) $g(t) = 16(1.02)^{\frac{8t}{5}}$

D) $g(t) = 16(1.02)^{\frac{5t}{8}}$

Answer

The data increase by 2% every 15 hours. In t days, there are $24t$ hours, so the number of 15-hour periods is:

$$\frac{24t}{15} = \frac{8t}{5}$$

Therefore:

$$g(t) = 16(1.02)^{\frac{8t}{5}}$$

Correct answer: C

Question

The exponential function g is defined by the given equation:

$$g(x) = 256(0.61)^{\frac{x}{30}}$$

The equation can be rewritten as:

$$g(x) = 256 \left(1 - \frac{k}{100} \right)^{\frac{x}{10}},$$

where k is a constant.

What is the value of k , to the nearest whole number?

- A) 15
- B) 39
- C) 61
- D) 77

Answer

We are given:

$$g(x) = 256(0.61)^{\frac{x}{30}} = 256 \left(1 - \frac{k}{100}\right)^{\frac{x}{10}}$$

Let:

$$r = 1 - \frac{k}{100} \Rightarrow \left(1 - \frac{k}{100}\right)^{\frac{x}{10}} = (0.61)^{\frac{x}{30}}$$

So:

$$r^{\frac{x}{10}} = (0.61)^{\frac{x}{30}} \Rightarrow r^{1/10} = (0.61)^{1/30}$$

Raise both sides to the 10th power:

$$r = \left((0.61)^{1/30}\right)^{10} = (0.61)^{\frac{1}{3}}$$

Now compute $(0.61)^{1/3} \approx 0.850$ (cube root of 0.61)

So:

$$1 - \frac{k}{100} \approx 0.85 \Rightarrow \frac{k}{100} \approx 0.15 \Rightarrow k \approx 15$$

Correct answer: A

Question

The function

$$W(t) = 830(1.09)^{\frac{6}{8}t}$$

gives the value, in dollars, of a certain investment after t years.

If the value of the investment increases by 9% every n months, what is the value of n ?

Answer

We are given:

$$W(t) = 830(1.09)^{\frac{6}{8}t}$$

The investment increases by 9% each time the exponent increases by 1.
Let m be the number of months. Since $t = \frac{m}{12}$:

$$\frac{6}{8}t = \frac{6}{8} \cdot \frac{m}{12} = \frac{m}{16}$$

So the value grows by 9% every 16 months.

16

Final Answer: $n = 16$

Question

To examine how a certain virus spreads, scientists introduced the virus to cells in a test tube and found that the number of infected cells in the test tube grew exponentially over time.

The given function C models the number of infected cells in the test tube t days after the virus was introduced:

$$C(t) = 100(2)^{\frac{t}{5}}$$

Which is the best interpretation of $(2)^{\frac{t}{5}}$ in this context?

- A) The predicted number of infected cells in the test tube doubled every 5 days.
- B) The predicted number of infected cells in the test tube grew by a factor of 5 every two days.
- C) The predicted number of infected cells in the test tube doubled every day.
- D) The predicted number of infected cells in the test tube grew by a factor of 5 every day.

Answer

The function is:

$$C(t) = 100(2)^{\frac{t}{5}}$$

This is an exponential growth function with base 2 and exponent $\frac{t}{5}$, which means the quantity doubles every 5 days. That's because:

$$C(5) = 100(2)^1 = 200 \Rightarrow \text{doubling in 5 days}$$

Correct answer: A

Question

The function f is defined for all $x \geq 0$.

Which of the following equivalent forms of the function f displays, as a constant or coefficient, the minimum value of f , where $x \geq 0$?

- A) $f(x) = 50(2.5)(1.25)^{x-2}$
- B) $f(x) = 125(0.8)(1.25)^{x-1}$
- C) $f(x) = 64(1.25)^{x+1}$
- D) $f(x) = 80(0.64)(1.25)^{x+2}$

Answer

We are asked to identify the expression that makes the **minimum value of** $f(x)$ appear as a constant or coefficient, given that $x \geq 0$.

For an exponential function in the form:

$$f(x) = A \cdot B^{x+c}$$

The minimum occurs when the exponent is minimized (i.e., $x = 0$).

Let's test $x = 0$ in all options:

A) $f(0) = 50(2.5)(1.25)^{-2} = 125 \cdot (1/1.5625) \approx 80$ **B)**

$f(0) = 125(0.8)(1.25)^{-1} = 100 \cdot (1/1.25) = 80$ **C)** $f(0) = 64(1.25)^1 = 80$ **D)**

$f(0) = 80(0.64)(1.25)^2 = 80(0.64)(1.5625) = 80$

All give the same result for $x = 0$: $f(0) = 80$. But the key is: which one displays the **minimum value directly as a constant or coefficient**?

Only **option D** has:

$$f(x) = \boxed{80}(0.64)(1.25)^{x+2}$$

and no exponent adjustment on the 80 — it's clearly the base value when $x = 0$.

Correct answer: D

Thank You!

We appreciate your time and focus.

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